Project Requirement Document

Amazon India E-Commerce Sales Analytics
Data Analyst Capstone | MySQL + Python + Power BI

Project title	End-to-End Sales Analytics for an Amazon India style E-Commerce Business
Prepared by	QuadSync Tech Solutions (Training Division)
Your role	Junior Data Analyst, working for the Business Intelligence team
Stakeholder	Head of Category and Growth (the person who will read your final dashboard)
Tools	MySQL Workbench, Python 3.10 (VS Code) with NumPy, Pandas, Matplotlib, Seaborn, Power BI Desktop
Duration	3 sessions, 6 stages (2 stages per session)

1. Business Background

An online marketplace sells products across eight categories to customers in 24 Indian cities. Leadership feels that sales are growing, but nobody can say which categories really make money, how much discounting is hurting margins, why returns are rising, or which customers are worth retaining. The raw data was exported from several internal systems and has **not been cleaned**. Your job is to turn it into decisions.

2. Project Objectives

- Build a reliable, cleaned analytical dataset from four raw tables.
- Answer the business questions in Section 6 using SQL, then explore deeper with Python.
- Deliver an interactive Power BI report that a non-technical manager can use on their own.
- Document every data quality problem you find and how you fixed it (the Data Quality Log).

3. Dataset Provided

Four CSV files (UTF-8). The file **orders.csv** is the main (fact) table with more than 165,000 rows. One row = one order line. **Warning: all files are raw exports. Do not assume any column is clean.**

File	Role	Approx. rows	Primary key	Links to
orders.csv	Fact (sales)	167,600	order_id	customer_id, product_id
customers.csv	Dimension	25,450	customer_id	orders.customer_id
products.csv	Dimension	1,535	product_id	orders.product_id
returns.csv	Fact (returns)	14,356	return_id	orders.order_id

Relationship: customers 1 --- * orders * --- 1 products ; orders 1 --- 0..1 returns (via order_id).

Data Dictionary

Table	Column	Meaning
orders	order_id	Unique order line identifier
orders	order_date / ship_date	Date the order was placed / dispatched
orders	customer_id, product_id	Foreign keys to customers and products

Table	Column	Meaning
orders	quantity, unit_price	Units bought; selling price per unit in INR
orders	discount_pct	Discount given, in percent
orders	order_total	Amount paid in INR (should equal quantity x unit_price x (1 - discount))
orders	payment_method, shipping_mode	How the customer paid / how the order was shipped
orders	order_status	Delivered, Shipped, Pending, Cancelled or Returned
orders	delivery_rating	Customer rating of delivery, 1 to 5
customers	customer_name, email, phone, gender	Customer profile and contact details
customers	signup_date, city, state, membership	Registration date, location, Standard or Prime
products	product_name, category, sub_category, brand	Product catalogue attributes
products	mrp, rating, stock_qty	List price in INR, average rating (1 to 5), units in stock
returns	return_id, order_id, return_date	Return record, original order, date returned
returns	return_reason, refund_amount, refund_status	Why returned, money refunded in INR, refund state

4. Project Workflow: 6 Stages over 3 Sessions

Work in the order below. Each stage has a deliverable, and the output of one stage is the input of the next.

Session	Stage	Tool	What you do	Deliverable
1	1. Understand and Load	MySQL	Read this document, create the database, load the 4 CSVs into staging tables, profile the raw data (row counts, NULLs, duplicates, odd values).	Loaded database + Data Quality Log v1
1	2. SQL Cleaning and Analysis	MySQL	Remove duplicates, trim and standardise text, treat blanks, flag orphan keys, build clean tables, then answer the SQL business questions.	clean_ tables + SQL query file
2	3. Python Cleaning	Python (Pandas, NumPy)	Connect to or read the clean tables; convert dates and prices to proper types; handle invalid numbers, outliers, missing values; validate totals; create new columns.	Cleaned CSV files + notebook
2	4. EDA and Insights	Python (Matplotlib, Seaborn)	Explore trends, distributions, correlations and segments with charts. Write 5 or more insights in plain English.	EDA notebook + final_dataset.csv
3	5. Data Model and DAX	Power BI	Import clean data, build a star schema, create a Date table, write DAX measures.	Model with relationships + measures
3	6. Dashboard and Story	Power BI	Design a 3-page report, add slicers, write insights and recommendations, present.	.pbix file + 3-minute presentation

5. Data Cleaning Requirements

You are expected to discover the problems yourself. As a starting point, check every table for the following categories, and record each finding (what, how many rows, how you fixed it) in the Data Quality Log:

- **Duplicates** - repeated rows or repeated keys.
- **Missing values** - NULLs, empty strings, and placeholders such as 'NA' or 'Unknown'.
- **Inconsistent text** - spelling variants, upper/lower case, extra spaces, abbreviations, alternative names.
- **Wrong data types or formats** - dates in different formats, numbers stored with symbols or percent signs, phone formats.
- **Invalid or impossible values** - negative or zero quantities, out-of-range ratings, discounts above 100 percent, dates in the wrong order.
- **Outliers** - values far outside the normal range for that column.
- **Referential integrity** - keys in one table that do not exist in the parent table.
- **Calculation mismatches** - totals that do not agree with their components.

Decision rule: for every problem choose one action - *fix, impute, flag, or remove* - and justify it. Never delete data silently.

6. Business Questions to Answer

SQL (Stage 2)

- What are total revenue, total orders, unique customers and average order value for each year?
- Which top 10 products and which categories generate the most revenue?
- What is the monthly revenue trend, and what is the month-over-month growth?
- Which cities and states contribute the most revenue? Rank them using a window function.
- Which payment methods are used most, and how does cancellation rate differ by payment method?
- Who are the top 20 customers by lifetime spend? How many customers ordered more than once (repeat rate)?
- What is the return rate by category, and what are the top return reasons?
- How does the average discount vary by category, and what is revenue share of orders discounted 30 percent or more?

Python (Stage 3 and 4)

- Document missing values before and after cleaning (Pandas summary table).
- Plot monthly and weekly revenue and identify the festive-season peak (Matplotlib line chart).
- Show the distribution of order value and detect outliers (Seaborn histogram and boxplot).
- Compare categories by revenue and return rate (Seaborn barplot) and examine the relationship between discount and quantity (scatter or heatmap).
- Compute a simple customer segmentation, e.g. frequency and spend buckets, using NumPy and Pandas (groupby, cut, where).
- Does delivery speed (ship date minus order date) affect delivery rating?

Power BI (Stage 5 and 6)

- **Page 1 - Executive Summary:** KPI cards (Revenue, Orders, Customers, AOV, Return Rate), monthly trend line, revenue by category.
- **Page 2 - Product and Customer Analysis:** top products, city map or bar, Prime vs Standard, payment method mix.
- **Page 3 - Returns and Operations:** returns by reason and category, delivery performance, cancellations.
- Slicers for Year, Category and State. Year-over-year comparison with a DAX measure. Consistent colours and clear titles.

7. Final Deliverables

#	Deliverable	Format
1	SQL scripts (load, cleaning, analysis queries)	.sql file(s)
2	Python notebook(s) with cleaning, EDA and written insights	.ipynb (VS Code)
3	Cleaned dataset	.csv files
4	Power BI report (3 pages)	.pbix
5	Data Quality Log (issue, count, action, reason)	.xlsx or .docx
6	3-minute presentation: problem, approach, 3 insights, 2 recommendations	Live demo

8. Evaluation Rubric (100 marks)

Area	Marks	What is checked
Data profiling and Data Quality Log	15	Problems found, counted and documented
SQL (cleaning and queries)	20	Correct joins, aggregation, window functions, readable code
Python cleaning	15	Correct types, justified handling of missing values and outliers
EDA and visualisation	15	Right chart for the question, labelled, insight written
Power BI model and DAX	15	Star schema, Date table, correct measures
Dashboard design and storytelling	15	Clarity, interactivity, actionable recommendations
Presentation	5	Clear, within time

9. Tips and Ground Rules

- Keep the raw staging tables and raw CSVs untouched. Always clean into new tables or files so your work is repeatable.
- Check row counts before and after every cleaning step.
- Validate your numbers across tools: total revenue in SQL, Python and Power BI should reconcile, and every difference must be explainable.
- Comment your code. Name files and columns consistently (snake_case).
- Business insight matters more than chart count. For every visual ask: what decision would this help?